Exercise - 2.6

1. Find the zeros of the polynomial function $f(x) = 4x^2 - 25$.
2. If $x = -2$ is one root of $x^3 - x^2 - 17x = 22$, then find the other roots of equation.
3. Find the real roots of $x^4 = 16$.
4. Solve $(2x + 1)^2 - (3x + 2)^2 = 0$.

Method of Undetermined Coefficients

Now let us focus on constructing polynomials with the given information using the method of undetermined coefficients. That is, we shall determine coefficients of the required polynomial using the given conditions. The main idea here is that two polynomials are equal if and only if the coefficients of same powers of the variables in the two polynomials are equal.

Example 2.16 Find a quadratic polynomial $f(x)$ such that, $f(0) = 1, f(-2) = 0$ and $f(1) = 0$.

Solution:
Let $f(x) = ax^2 + bx + c$ be the polynomial satisfying the given conditions.
$f(0) = a(0)^2 + b(0) + c = 1$, implies that $c = 1$. Now the other two conditions $f(-2) = 0$, $f(1) = 0$ give $4a - 2b + c = 0$ and $a + b + c = 0$.
Using $c = 1$, we get $4a - 2b = -1$ and $a + b = -1$. Solving these two equations we get $a = b = -\frac{1}{2}$ and thus, we have $f(x) = -\frac{1}{2}x^2 - \frac{1}{2}x + 1$.

The above problem can also be solved in another way. $x = -2$, $x = 1$ are zeros of $f(x)$. Thus, $f(x) = d(x + 2)(x - 1)$ for some constant d.
Now using $f(0) = 1$ gives $-2d = 1$, hence $d = -\frac{1}{2}$. So, $f(x) = -\frac{1}{2}(x+2)(x-1) = -\frac{1}{2}x^2 - \frac{1}{2}x + 1$.

Example 2.17 Construct a cubic polynomial function with rational coefficients having zeros at $x = \frac{2}{5}, 1 + \sqrt{3}$ such that $f(0) = -8$.

Solution:
Given that $\frac{2}{5}$ and $1 + \sqrt{3}$ are zeros of $f(x)$. Thus, $1 - \sqrt{3}$ is also a zero of $f(x)$.
Let $f(x) = a(x - \frac{2}{5})[(x - (1 + \sqrt{3}))][x - (1 - \sqrt{3})] = a(x - \frac{2}{5})[(x - 1)^2 - 3]$.
Using $f(0) = -8$, we have, $\left(-\frac{2}{5}a\right)(-2) = -8$ which give $a = -10$.
Thus the required polynomial is $f(x) = (-10)(x - \frac{2}{5})[x^2 - 2x - 2] = -10x^3 + 24x^2 + 12x - 8$.

Example 2.18 Prove that $ap + q = 0$ if $f(x) = x^3 - 3px + 2q$ is divisible by $g(x) = x^2 + 2ax + a^2$.

Solution:
Note that the degree of $f(x)$ is 3 and the leading coefficient is 1. Since $g(x)$ divides $f(x)$, we have $f(x) = (x + b)g(x)$, for some $b \in \mathbb{R}$. Thus, $x^3 - 3px + 2q = (x + b)(x^2 + 2ax + a^2)$.
Equating like coefficients on both sides, we have $2a + b = 0$, $a^2 + 2ab = -3p$ and $2q = ba^2$. Thus, $b = -2a$, $p = a^2$, and $q = -a^3$.
Now, $q = -a^3 = -a(a^2) = -ap$, which gives $ap + q = 0$.